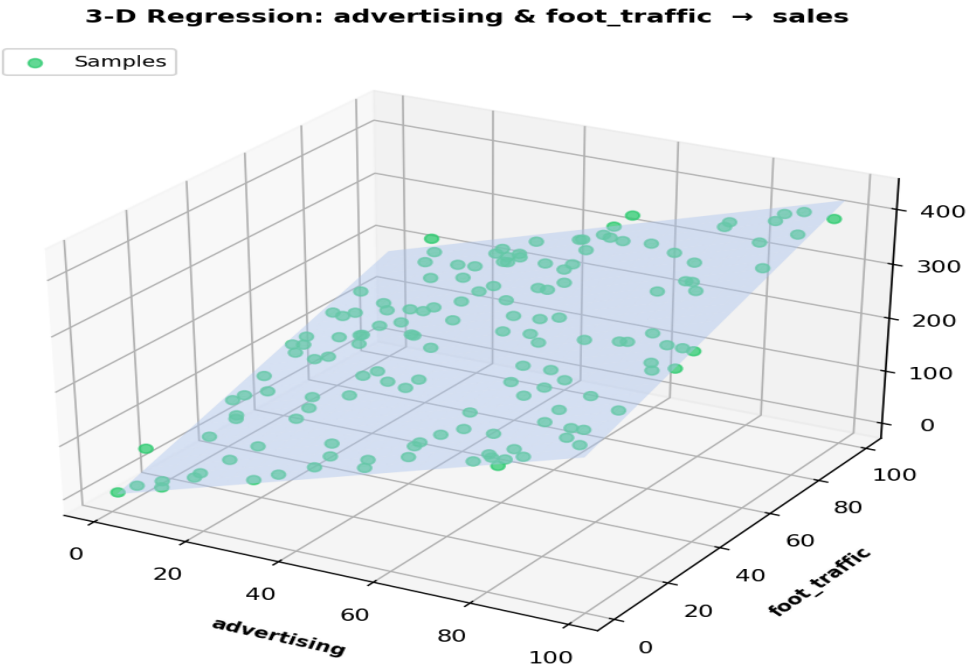
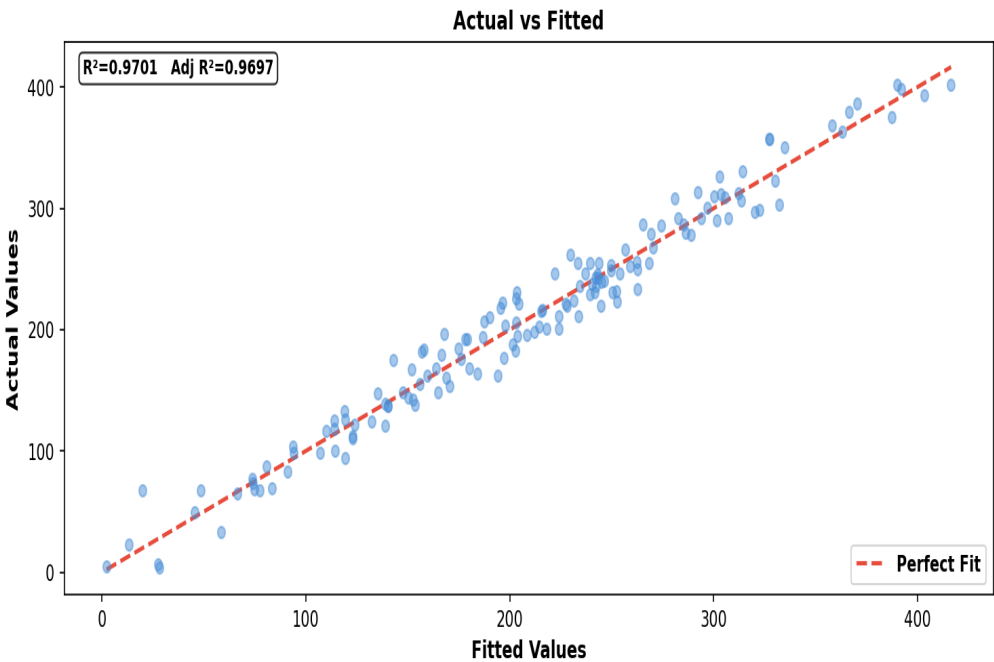


OLS Regression - Diagnostic Audit Report

Target: sales | Predictors: advertising, foot_traffic | Best Model: Linear | n = 150

OLS Regression Plot (Linear) & 3-D Scatter + Regression Plane



Diagnostic Scorecard - Best Model (Linear)

Test	Statistic	p-value	Threshold	Result	Notes
F-Test (Overall)	2385.6584	0.0000	$p < 0.05$	PASS	df model=2, resid=147
Normality - Shapiro-Wilk	0.9882	0.2346	$p > 0.05$	PASS	Tests residual normality
Normality - Jarque-Bera	2.5783	0.2755	$p > 0.05$	PASS	Skew=0.270, Kurt=2.653
Homoscedasticity - Breusch-Pagan	0.7694	0.6806	$p > 0.05$	PASS	Tests constant variance
Autocorrelation - Durbin-Watson	1.8108	-	1.5 – 2.5	PASS	Tests residual independence
Multicollinearity - Max VIF	1.01	-	< 5	PASS	Predictors: advertising, foot_traffic
Influential Obs - Cook's D	6 flagged	-	$4/n = 0.0267$	FAIL	Red circles on residual plot

OLS Regression - Interpretation & Model Comparison

Target: sales | Predictors: advertising, foot_traffic | Best Model: Linear | n = 150

Interpretation of Results

Overall Fit	The best model (degree 1) explains 97.0% of variance in sales ($R^2 = 0.9701$, Adj $R^2 = 0.9697$), indicating a strong fit.
Model Significance (F-Test)	$F(2, 147) = 2385.658$, $p = 0.0000$. The overall regression is statistically significant ($p < 0.05$).
Normality of Residuals (Shapiro-Wilk)	$W = 0.9882$, $p = 0.2346$. Residuals appear normally distributed — the normality assumption is met.
Homoscedasticity (Breusch-Pagan)	BP stat = 0.7694, $p = 0.6806$. Residual variance appears constant (homoscedastic).
Autocorrelation (Durbin-Watson)	DW = 1.8108. No significant autocorrelation detected.
Multicollinearity (VIF)	Max VIF = 1.01. No multicollinearity concern (all VIF < 5).
Influential Observations (Cook's Distance)	6 influential observation(s) detected (Cook's D > 0.0267). Review flagged points for data entry errors or genuine outliers.

Top 5 Models Ranked by BIC

Models ranked by original-scale BIC — lower = better fit penalised for complexity. SW p = Shapiro-Wilk (normality); BP p = Breusch-Pagan (homoscedasticity). Green = PASS ($p > 0.05$), Red = FAIL ($p \leq 0.05$). Single-predictor runs also test Logarithmic, Exponential, Log-Log, Reciprocal, Spline, and Lag-1 variants, producing up to 10 candidates. Multi-predictor runs fit Linear and Interaction models only, so fewer rows appear.

Rank	Model Type	Fitted Equation	R^2	Adj R^2	AIC	BIC	SW p	BP p
1	Linear	sales = -1.743 + 2.4886*advertising + 1.8174*foot_traffic	0.9701	0.9697	1254.01	1263.04	0.2346	0.6806
2	Polynomial (deg 2)	sales = -1.743 + 2.4886*advertising + 1.8174*foot_traffic	0.9701	0.9697	1254.01	1263.04	0.2346	0.6806
3	Polynomial (deg 3)	sales = -1.743 + 2.4886*advertising + 1.8174*foot_traffic	0.9701	0.9697	1254.01	1263.04	0.2346	0.6806
4	Polynomial (deg 4)	sales = -1.743 + 2.4886*advertising + 1.8174*foot_traffic	0.9701	0.9697	1254.01	1263.04	0.2346	0.6806
5	Interaction	sales = -0.304 + 2.4606*advertising + 1.7878*foot_traffic + 0.0006*advertising*foot_traffic	0.9701	0.9695	1255.84	1267.88	0.2425	0.8582

Diagnostic Plots - Best Model

OLS Diagnostic Plots

